

34721/34722 Linear Control Design 1

Spring Semester

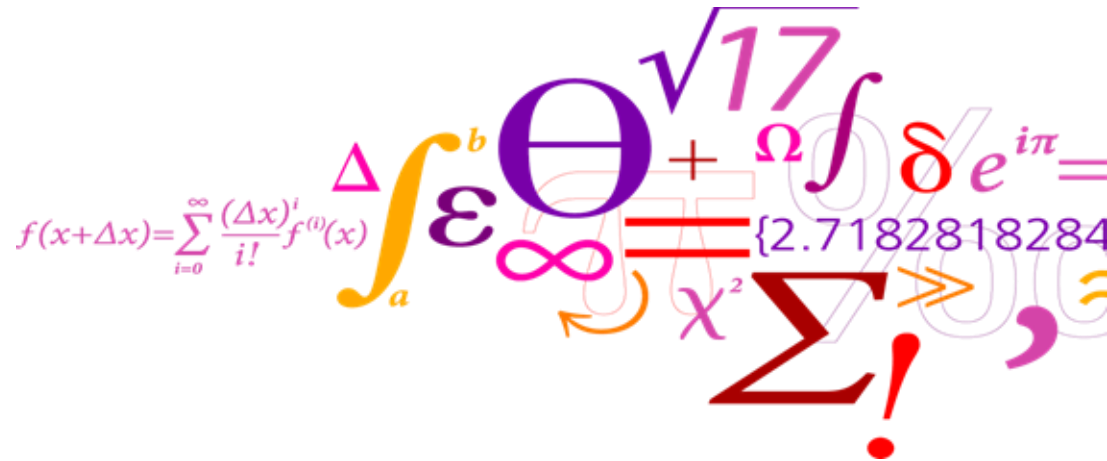
Silvia Tolu - Dimitrios Papageorgiou



Lecture 12/13

Schedule (13:00-15:00)

- **Disturbances**
- **Sensitivity**
- **Pre-filters**



What you will learn today

After this lecture you will be able to

- Find the transfer function from a disturbance to the output.
- Calculate the steady-state error due to a disturbance.
- Explain the sensitivity of system output and control input to disturbances.
- Explain the use of pre-filters in the reference
- Design first and second-order pre-filters

Disturbances

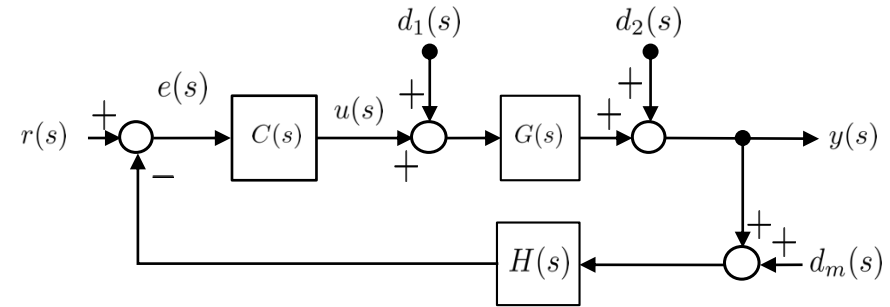
Some facts:

- A disturbance is an unwanted (usually unknown) input.
- In a linear system with multiple inputs/disturbances, each input contributes **separately** to the output (superposition principle).
- We can find the transfer function from an input to an output by setting all the other inputs to 0.
- The total output is:

$$y(s) = G_{yr}(s)r(s) + G_{yd_1}(s)d_1(s) + G_{yd_2}(s)d_2(s) + G_{yd_m}(s)d_m(s)$$

- The total error is:

$$e(s) = G_{er}(s)r(s) + G_{ed_1}(s)d_1(s) + G_{ed_2}(s)d_2(s) + G_{ed_m}(s)d_m(s)$$



$$G_{yr}(s) = \frac{C(s)G(s)}{1 + C(s)G(s)H(s)}$$

$$G_{yd_1}(s) = \frac{G(s)}{1 + C(s)G(s)H(s)}$$

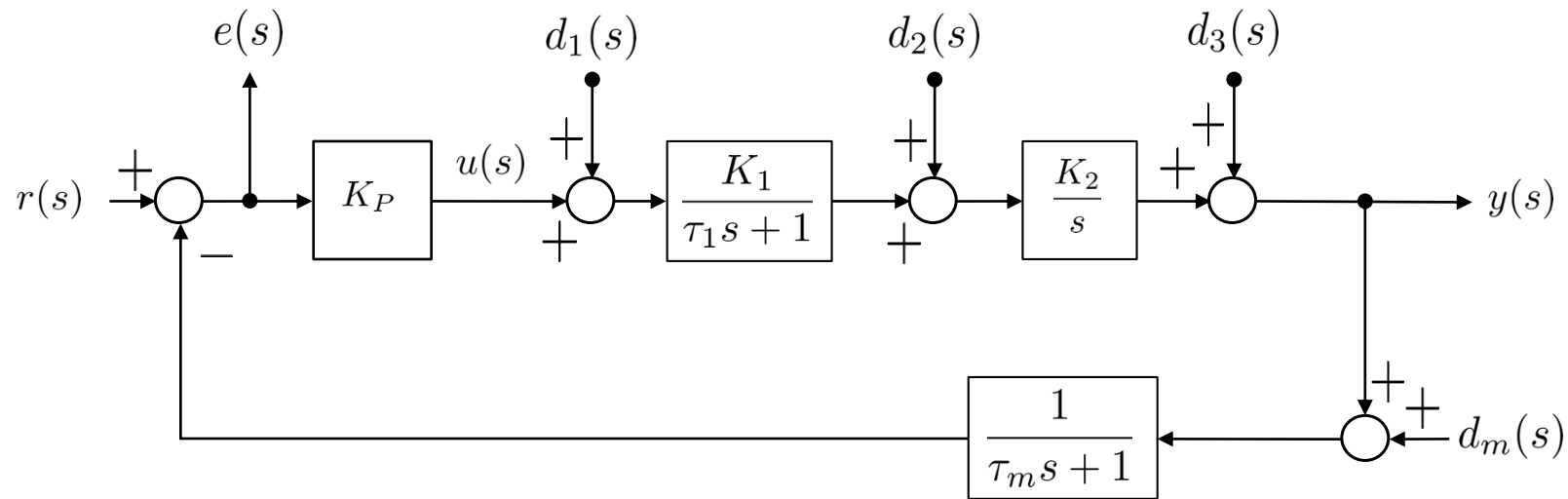
$$G_{yd_2}(s) = \frac{1}{1 + C(s)G(s)H(s)}$$

$$G_{yd_m}(s) = \frac{-C(s)G(s)H(s)}{1 + C(s)G(s)H(s)}$$

Disturbances

Example:

For the given system, find the transfer functions from r , d_1 , d_2 , d_3 and d_m to the output y and error e .



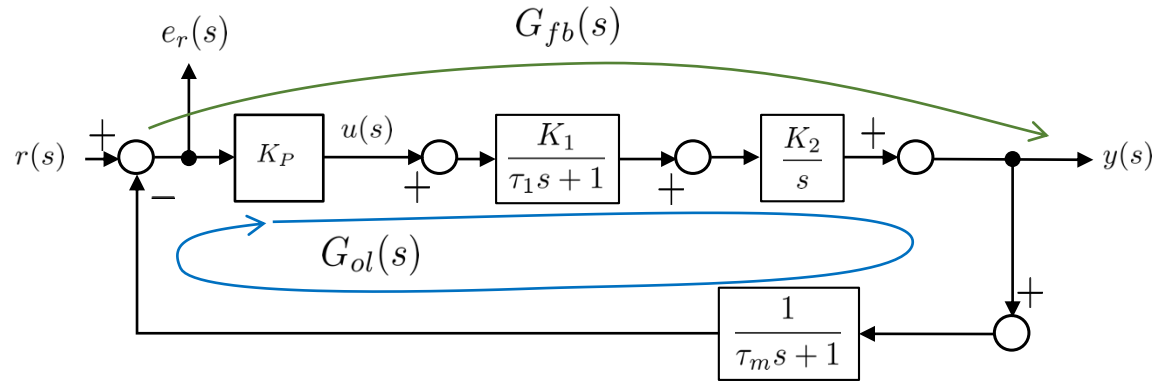
Disturbances

Example: Find the transfer functions from r , d_1 , d_2 , d_3 and d_m to the output y and error e .

- From r to y ($d_1 = d_2 = d_3 = d_m = 0$)

$$G_{fb}(s) = \frac{K_P K_1 K_2}{s(\tau_1 s + 1)}$$

$$G_{ol}(s) = \frac{K_P K_1 K_2}{s(\tau_1 s + 1)(\tau_m s + 1)}$$



$$G_{yr}(s) = \frac{G_{fb}(s)}{1 + G_{ol}(s)} = \frac{K_P \frac{K_1}{\tau_1 s + 1} \cdot \frac{K_2}{s}}{1 + K_P \frac{K_1}{\tau_1 s + 1} \cdot \frac{K_2}{s} \cdot \frac{1}{\tau_m s + 1}} = \frac{(\tau_m s + 1) K_P K_1 K_2}{s(\tau_1 s + 1)(\tau_m s + 1) + K_P K_1 K_2}$$

$$e_r(s) = r(s) - \frac{1}{\tau_m s + 1} \cdot y(s) \Rightarrow G_{er}(s) = 1 - \frac{1}{\tau_m s + 1} G_{yr}(s) = 1 - \frac{1}{\tau_m s + 1} \cdot \frac{G_{fb}(s)}{1 + G_{ol}(s)} = 1 - \frac{G_{ol}(s)}{1 + G_{ol}(s)} = \frac{1}{1 + G_{ol}(s)}$$

$$= \frac{s(\tau_1 s + 1)(\tau_m s + 1)}{s(\tau_1 s + 1)(\tau_m s + 1) + K_P K_1 K_2}$$

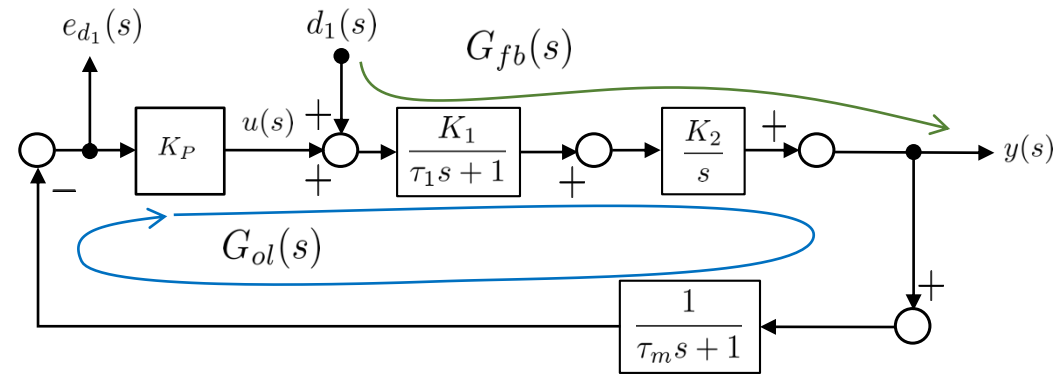
Disturbances

Example: Find the transfer functions from r , d_1 , d_2 , d_3 and d_m to the output y and error e .

- From d_1 to y ($r = d_2 = d_3 = d_m = 0$)

$$G_{fb}(s) = \frac{K_1 K_2}{s(\tau_1 s + 1)}$$

$$G_{ol}(s) = \frac{K_P K_1 K_2}{s(\tau_1 s + 1)(\tau_m s + 1)}$$



$$G_{yd_1}(s) = \frac{G_{fb}(s)}{1 + G_{ol}(s)} = \frac{\frac{K_1}{\tau_1 s + 1} \cdot \frac{K_2}{s}}{1 + K_P \frac{K_1}{\tau_1 s + 1} \cdot \frac{K_2}{s} \cdot \frac{1}{\tau_m s + 1}} = \frac{(\tau_m s + 1) K_1 K_2}{s(\tau_1 s + 1)(\tau_m s + 1) + K_P K_1 K_2}$$

$$e_{d_1}(s) = -\frac{K_1}{\tau_1 s + 1} \cdot \frac{K_2}{s} \cdot \frac{1}{\tau_m s + 1} \cdot (K_P e_{d_1}(s) + d_1(s)) \Rightarrow G_{ed_1}(s) = -\frac{K_1 K_2}{s(\tau_1 s + 1)(\tau_m s + 1)} \cdot \frac{1}{1 + G_{ol}(s)}$$

$$= -\frac{K_1 K_2}{s(\tau_1 s + 1)(\tau_m s + 1) + K_P K_1 K_2}$$

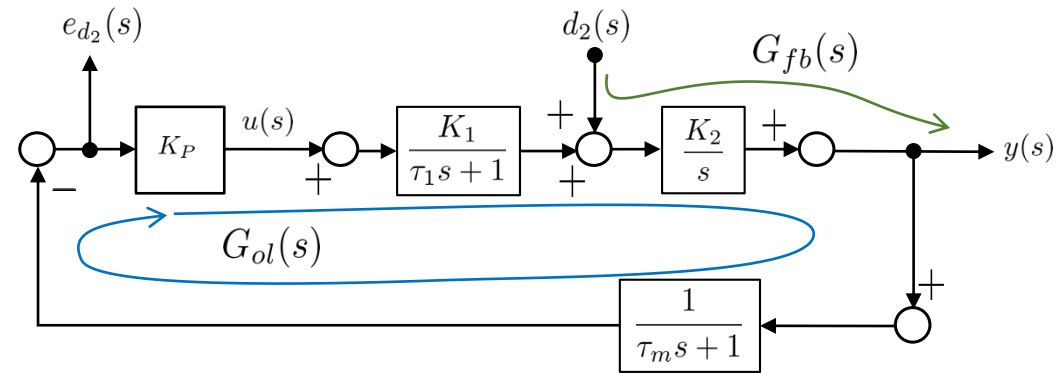
Disturbances

Example: Find the transfer functions from r , d_1 , d_2 , d_3 and d_m to the output y and error e .

- From d_2 to y ($r = d_1 = d_3 = d_m = 0$)

$$G_{fb}(s) = \frac{K_2}{s}$$

$$G_{ol}(s) = \frac{K_P K_1 K_2}{s(\tau_1 s + 1)(\tau_m s + 1)}$$



$$G_{yd_2}(s) = \frac{G_{fb}(s)}{1 + G_{ol}(s)} = \frac{\frac{K_2}{s}}{1 + K_P \frac{K_1}{\tau_1 s + 1} \cdot \frac{K_2}{s} \cdot \frac{1}{\tau_m s + 1}} = \frac{(\tau_m s + 1)(\tau_1 s + 1)K_2}{s(\tau_1 s + 1)(\tau_m s + 1) + K_P K_1 K_2}$$

$$e_{d_2}(s) = -\frac{K_2}{s} \cdot \frac{1}{\tau_m s + 1} \cdot \left(\frac{K_P K_1}{\tau_1 s + 1} e_{d_2}(s) + d_2(s) \right) \Rightarrow G_{ed_2}(s) = -\frac{K_2}{s(\tau_m s + 1)} \cdot \frac{1}{1 + G_{ol}(s)}$$

$$= -\frac{K_2(\tau_1 s + 1)}{s(\tau_1 s + 1)(\tau_m s + 1) + K_P K_1 K_2}$$

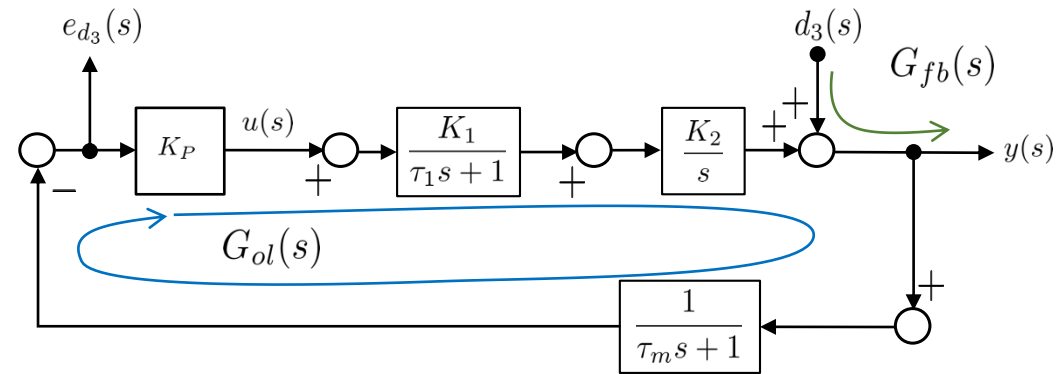
Disturbances

Example: Find the transfer functions from r , d_1 , d_2 , d_3 and d_m to the output y and error e .

- From d_3 to y ($r = d_1 = d_2 = d_m = 0$)

$$G_{fb}(s) = 1$$

$$G_{ol}(s) = \frac{K_P K_1 K_2}{s(\tau_1 s + 1)(\tau_m s + 1)}$$



$$G_{yd_3}(s) = \frac{G_{fb}(s)}{1 + G_{ol}(s)} = \frac{1}{1 + K_P \frac{K_1}{\tau_1 s + 1} \cdot \frac{K_2}{s} \cdot \frac{1}{\tau_m s + 1}} = \frac{s(\tau_m s + 1)(\tau_1 s + 1)}{s(\tau_1 s + 1)(\tau_m s + 1) + K_P K_1 K_2} (= G_{er}(s)!!!)$$

$$e_{d_3}(s) = -\frac{1}{\tau_m s + 1} \cdot \left(\frac{K_P K_1 K_2}{s(\tau_1 s + 1)} e_{d_3}(s) + d_3(s) \right) \Rightarrow G_{ed_3}(s) = -\frac{1}{\tau_m s + 1} \cdot \frac{1}{1 + G_{ol}(s)}$$

$$= -\frac{s(\tau_1 s + 1)}{s(\tau_1 s + 1)(\tau_m s + 1) + K_P K_1 K_2}$$

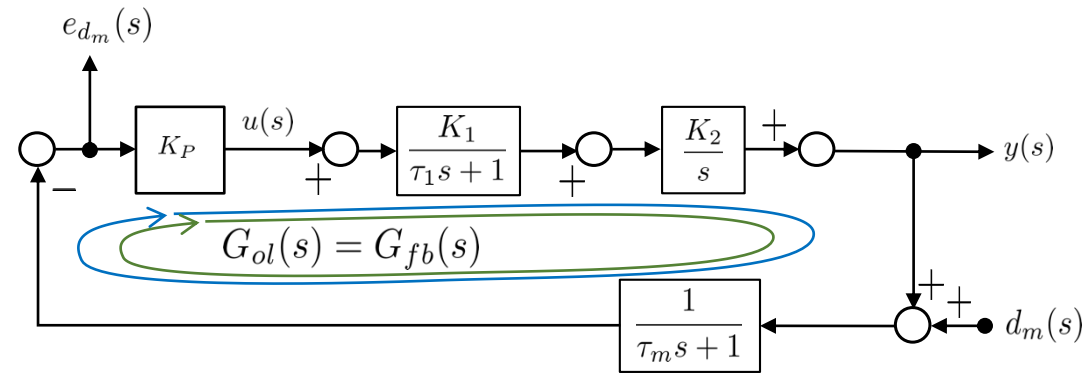
Disturbances

Example: Find the transfer functions from r , d_1 , d_2 , d_3 and d_m to the output y and error e .

- From d_m to y ($r = d_1 = d_2 = d_3 = 0$)

$$G_{fb}(s) = -\frac{K_P K_1 K_2}{s(\tau_1 s + 1)(\tau_m s + 1)}$$

$$G_{ol}(s) = \frac{K_P K_1 K_2}{s(\tau_1 s + 1)(\tau_m s + 1)}$$



$$G_{y d_m}(s) = \frac{G_{fb}(s)}{1 + G_{ol}(s)} = -\frac{K_P \frac{K_1}{\tau_1 s + 1} \cdot \frac{K_2}{s} \cdot \frac{1}{\tau_m s + 1}}{1 + K_P \frac{K_1}{\tau_1 s + 1} \cdot \frac{K_2}{s} \cdot \frac{1}{\tau_m s + 1}} = -\frac{K_P K_1 K_2}{s(\tau_1 s + 1)(\tau_m s + 1) + K_P K_1 K_2}$$

$$e_{d_m}(s) = -\frac{1}{\tau_m s + 1} \cdot \left(\frac{K_P K_1 K_2}{s(\tau_1 s + 1)} e_{d_m}(s) + d_m(s) \right) \Rightarrow G_{e d_m}(s) = -\frac{1}{\tau_m s + 1} \cdot \frac{1}{1 + G_{ol}(s)}$$

$$= -\frac{s(\tau_1 s + 1)}{s(\tau_1 s + 1)(\tau_m s + 1) + K_P K_1 K_2} (= G_{e d_3}(s)!!!)$$

Disturbances

Example:

What about the steady state errors for step changes in the disturbances and the reference?

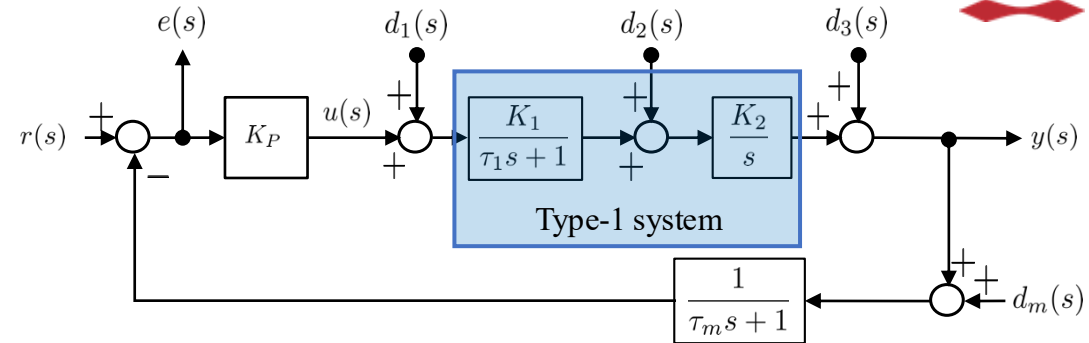
$$e_r(s) = \frac{s(\tau_1 s + 1)(\tau_m s + 1)}{s(\tau_1 s + 1)(\tau_m s + 1) + K_P K_1 K_2} r(s)$$

$$e_{d_1}(s) = -\frac{K_1 K_2}{s(\tau_1 s + 1)(\tau_m s + 1) + K_P K_1 K_2} d_1(s)$$

$$e_{d_2}(s) = -\frac{K_2(\tau_1 s + 1)}{s(\tau_1 s + 1)(\tau_m s + 1) + K_P K_1 K_2} d_2(s)$$

$$e_{d_3}(s) = -\frac{s(\tau_1 s + 1)}{s(\tau_1 s + 1)(\tau_m s + 1) + K_P K_1 K_2} d_3(s)$$

$$e_{d_m}(s) = -\frac{s(\tau_1 s + 1)}{s(\tau_1 s + 1)(\tau_m s + 1) + K_P K_1 K_2} d_m(s)$$



$$e_{r,ss} = 0$$

$$e_{d_1,ss} = -\frac{1}{K_P}$$

$$e_{d_2,ss} = -\frac{1}{K_P K_1}$$

$$e_{d_3,ss} = 0$$

$$e_{d_m,ss} = 0$$

$$e_{ss} = \lim_{s \rightarrow 0} s G_e(s) \frac{1}{s}$$



Final value theorem

Does this mean that the measurement disturbance gives no error?

Disturbances

Example:

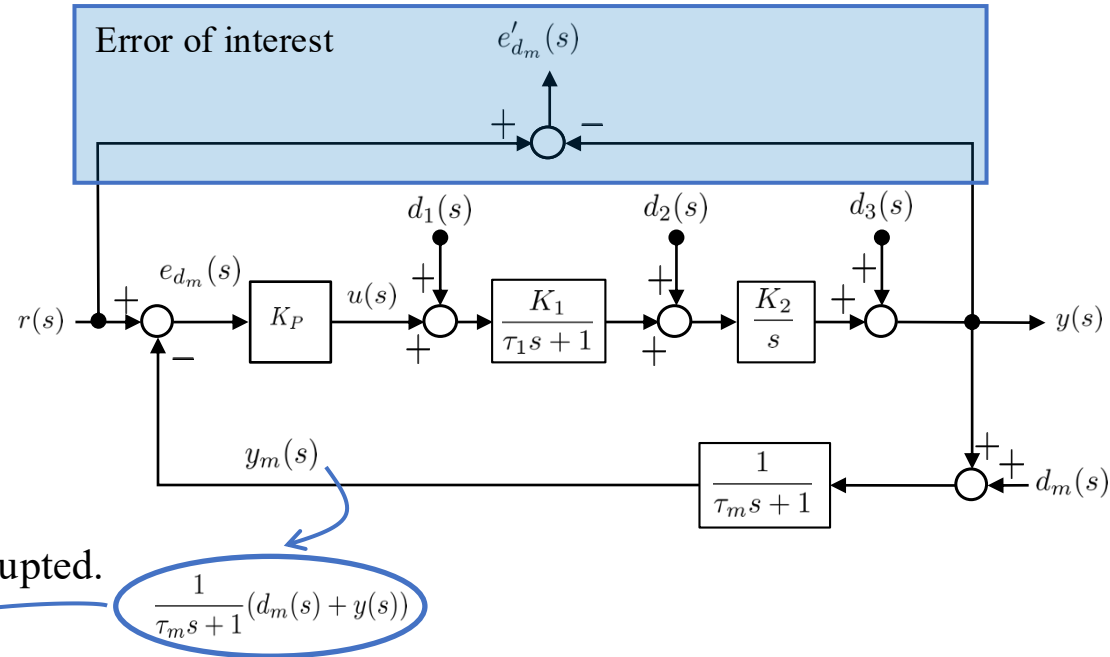
What about the steady state errors for step changes in the disturbances and the reference?

$$e_{d_m}(s) = -\frac{s(\tau_1 s + 1)}{s(\tau_1 s + 1)(\tau_m s + 1) + K_P K_1 K_2} d_m(s)$$

$$e_{d_m,ss} = 0$$

Does this mean that the measurement step disturbance gives no error?

- This error is not suitable for performance evaluation: the measurements are already corrupted.
- Use $r(s) - y(s)$ instead



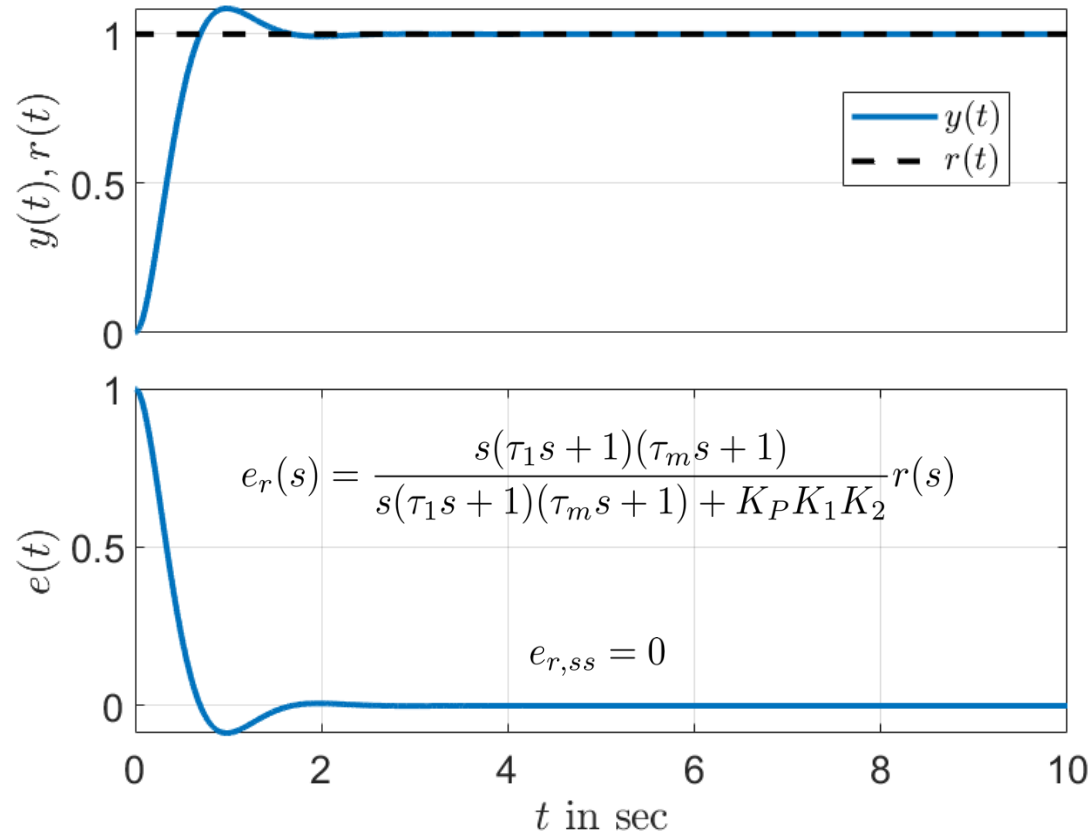
$$e'_{d_m}(s) = r(s) - y(s) = e_{d_m} + y_m(s) - y(s) = \left[G_{ed_m}(s) + \frac{1}{\tau_m s + 1} + \left(\frac{1}{\tau_m s + 1} - 1 \right) G_{yd_m}(s) \right] d_m(s) \Rightarrow$$

$$e'_{d_m,ss} = \lim_{s \rightarrow 0} s \left[G_{ed_m}(s) + \frac{1}{\tau_m s + 1} + \left(\frac{1}{\tau_m s + 1} - 1 \right) G_{yd_m}(s) \right] \frac{1}{s} = G_{ed_m}(0) + 1 \Rightarrow e'_{d_m,ss} = 1$$

Disturbances

steady_state_errors.m

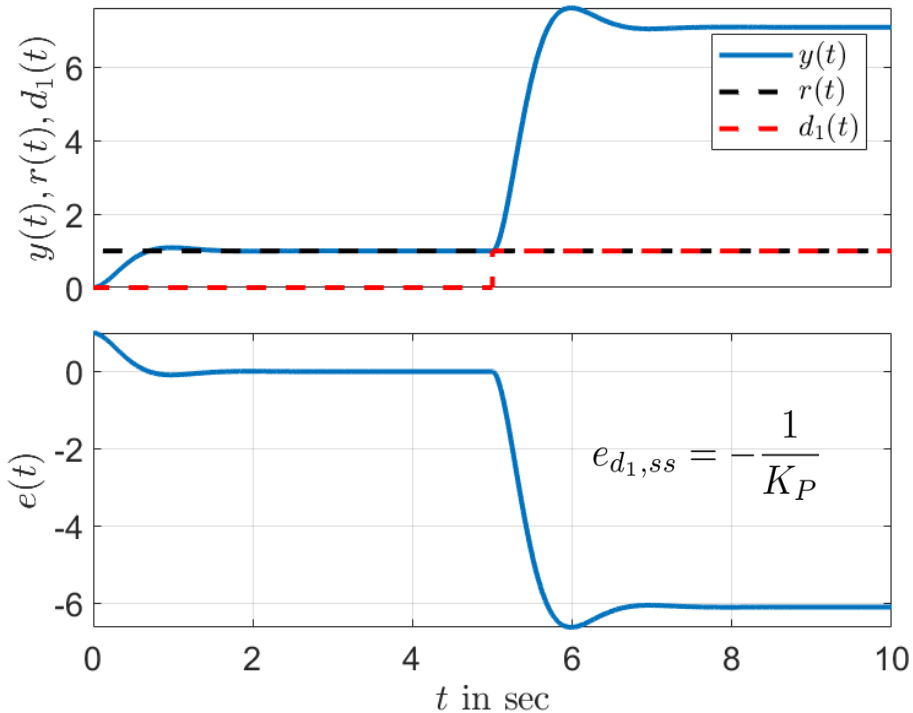
Example: Application in Matlab for step changes in the disturbances and constant reference?



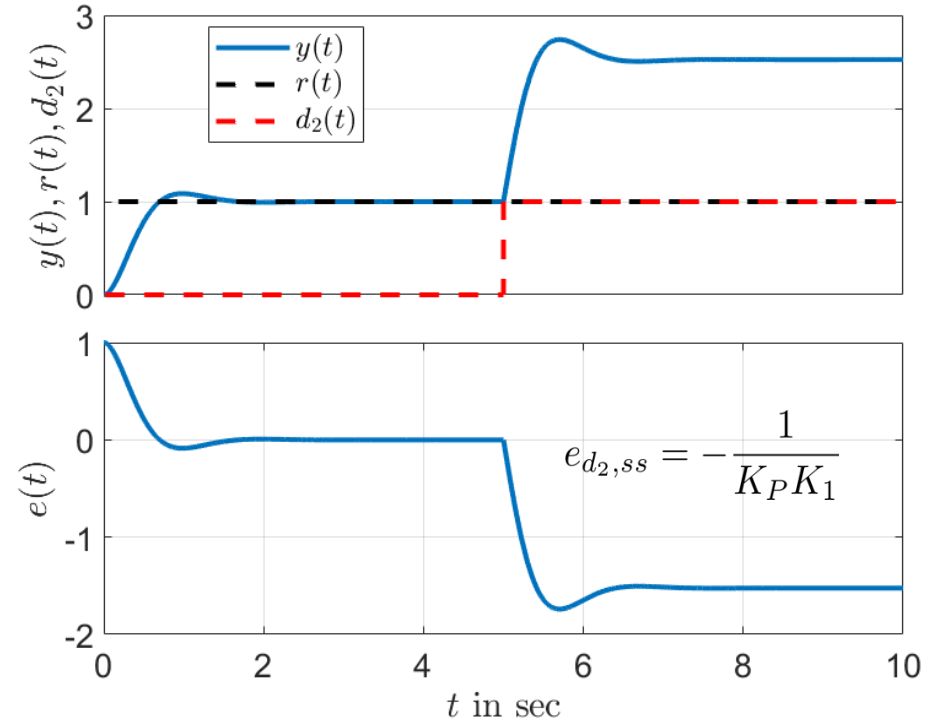
Disturbances

steady_state_errors.m

Example: Application in Matlab for step changes in the disturbances and constant reference?



$$e_{d_1}(s) = -\frac{K_1 K_2}{s(\tau_1 s + 1)(\tau_m s + 1) + K_P K_1 K_2} d_1(s)$$

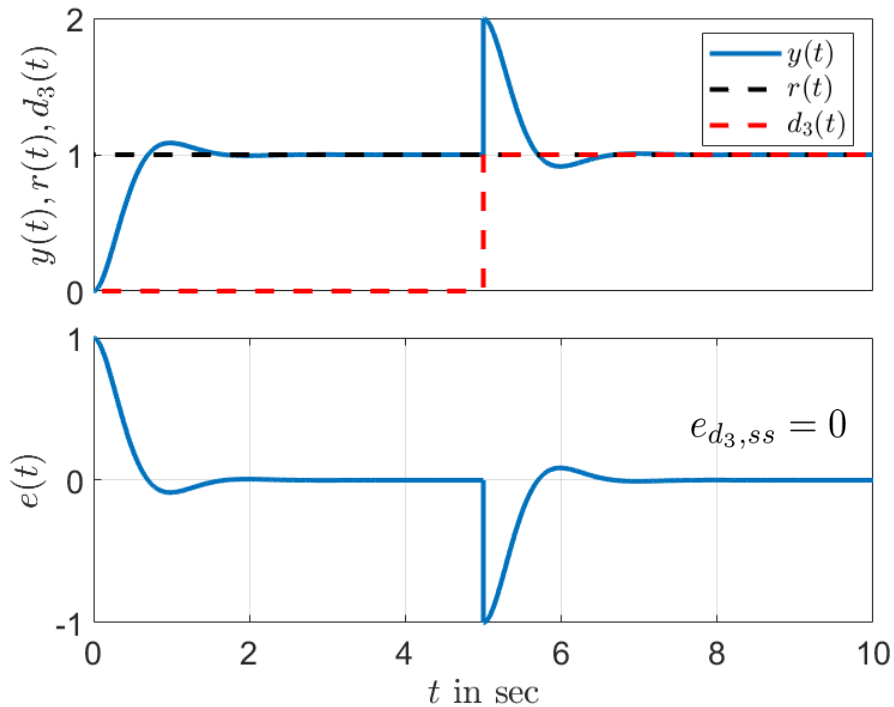


$$e_{d_2}(s) = -\frac{K_2(\tau_1 s + 1)}{s(\tau_1 s + 1)(\tau_m s + 1) + K_P K_1 K_2} d_2(s)$$

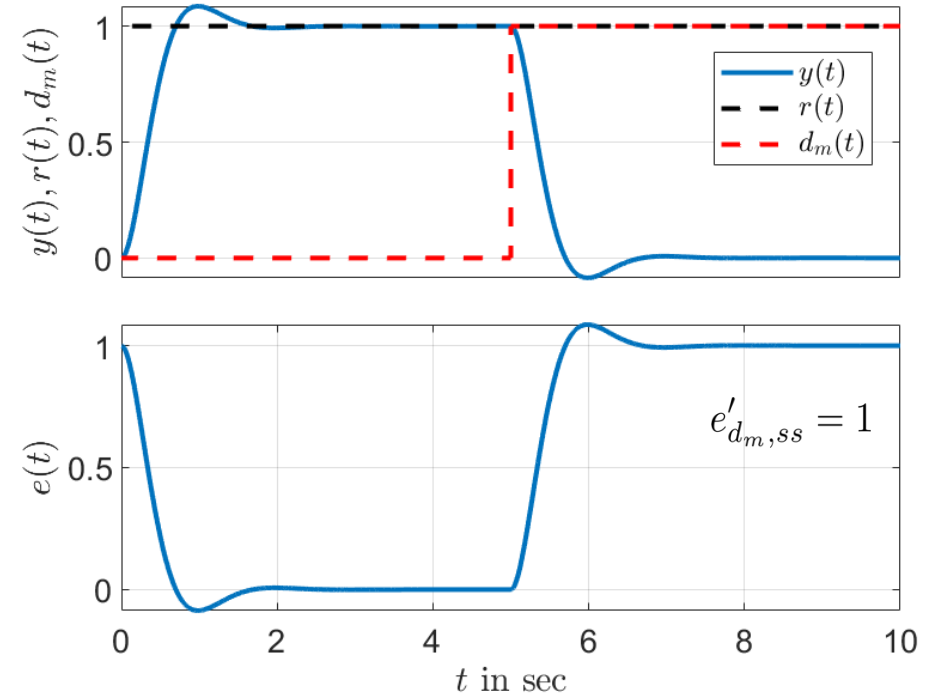
Disturbances

steady_state_errors.m

Example: Application in Matlab for step changes in the disturbances and constant reference?



$$e_{d_3}(s) = -\frac{s(\tau_1 s + 1)}{s(\tau_1 s + 1)(\tau_m s + 1) + K_P K_1 K_2} d_3(s)$$

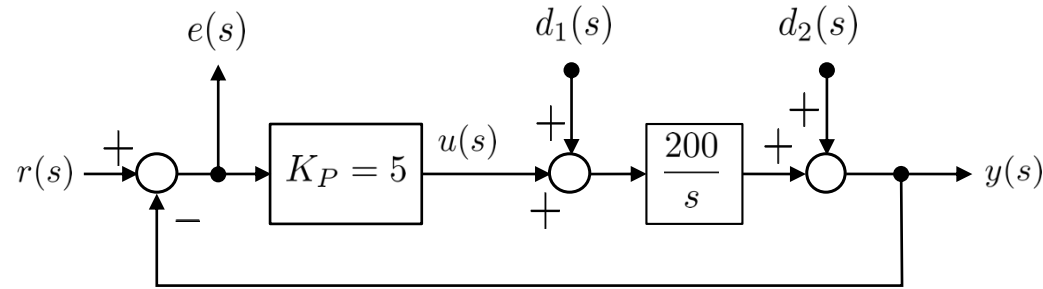


$$e_{d_m}(s) = -\frac{s(\tau_1 s + 1)}{s(\tau_1 s + 1)(\tau_m s + 1) + K_P K_1 K_2} d_m(s)$$

Disturbances

Exercise 1

disturbances_exercise_1.m



- Does a step change in r give non-zero steady-state error?

$$G_{er}(s) = \frac{1}{1 + 5\frac{200}{s}} = \frac{s}{s + 1000} \Rightarrow e_{r,ss} = \lim_{s \rightarrow 0} s G_{er}(s) \frac{1}{s} = \lim_{s \rightarrow 0} \frac{s}{s + 1000} = 0$$

- Does a step change in d_1 give non-zero steady-state error?

$$G_{ed_1}(s) = \frac{-\frac{200}{s}}{1 + 5\frac{200}{s}} = -\frac{200}{s + 1000} \Rightarrow e_{d_1,ss} = \lim_{s \rightarrow 0} s G_{ed_1}(s) \frac{1}{s} = \lim_{s \rightarrow 0} \frac{-200}{s + 1000} = -0.2$$

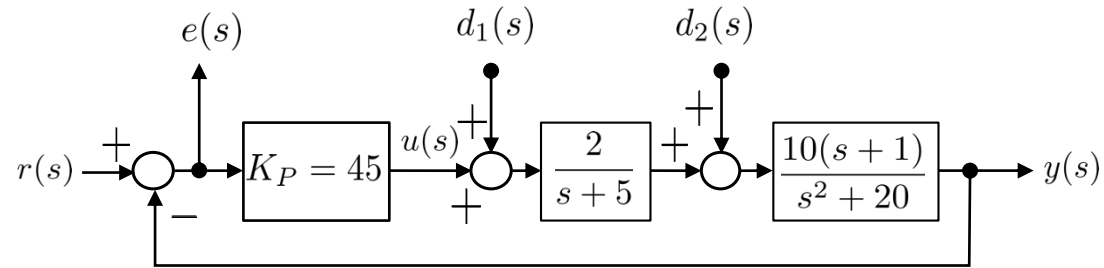
- Does a step change in d_2 give non-zero steady-state error?

$$G_{ed_2}(s) = \frac{-1}{1 + 5\frac{200}{s}} = -\frac{s}{s + 1000} \Rightarrow e_{d_2,ss} = \lim_{s \rightarrow 0} s G_{ed_2}(s) \frac{1}{s} = \lim_{s \rightarrow 0} \frac{-s}{s + 1000} = 0$$

Disturbances

Exercise 2

disturbances_exercise_2.m



- Does a step change in r give non-zero steady-state error?

$$G_{er}(s) = \frac{1}{1 + 45 \frac{20(s+1)}{(s+5)(s^2+20)}} = \frac{s^3 + 5s^2 + 20s + 100}{s^3 + 5s^2 + 920s + 1000} \Rightarrow e_{r,ss} = \lim_{s \rightarrow 0} s G_{er}(s) \frac{1}{s} = 0.1$$

- Does a step change in d_1 give non-zero steady-state error?

$$G_{ed_1}(s) = \frac{-\frac{20(s+1)}{(s+5)(s^2+20)}}{1 + 45 \frac{20(s+1)}{(s+5)(s^2+20)}} = \frac{-20(s+1)}{s^3 + 5s^2 + 920s + 1000} \Rightarrow e_{d_1,ss} = \lim_{s \rightarrow 0} s G_{ed_1}(s) \frac{1}{s} = -0.02$$

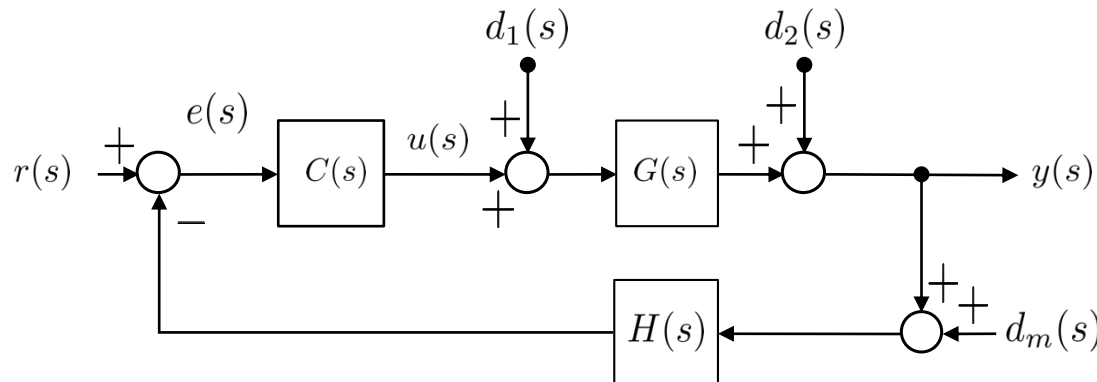
- Does a step change in d_2 give non-zero steady-state error?

$$G_{ed_2}(s) = \frac{-\frac{10(s+1)}{s^2+20}}{1 + 45 \frac{20(s+1)}{(s+5)(s^2+20)}} = \frac{-10s^2 - 60s - 50}{s^3 + 5s^2 + 920s + 1000} \Rightarrow e_{d_2,ss} = \lim_{s \rightarrow 0} s G_{ed_2}(s) \frac{1}{s} = -0.05$$

Sensitivity

- The effect of a disturbance on the output is important not just at steady-state but at other frequencies as well.
- The transfer function $G_{yd}(s)$ from each disturbance d to the output y is called the **sensitivity** of the output to the disturbance.

$$y(s) = \frac{C(s)G(s)}{1 + G_{ol}(s)} r(s) + \underbrace{\frac{G(s)}{1 + G_{ol}(s)}}_{\text{Sensitivity of } y(s) \text{ to } d_1(s)} d_1(s) + \underbrace{\frac{1}{1 + G_{ol}(s)}}_{\text{Sensitivity of } y(s) \text{ to } d_2(s)} d_2(s) + \underbrace{\frac{-G_{ol}(s)}{1 + G_{ol}(s)}}_{\text{Sensitivity of } y(s) \text{ to } d_m(s)} d_m(s)$$



Sensitivity

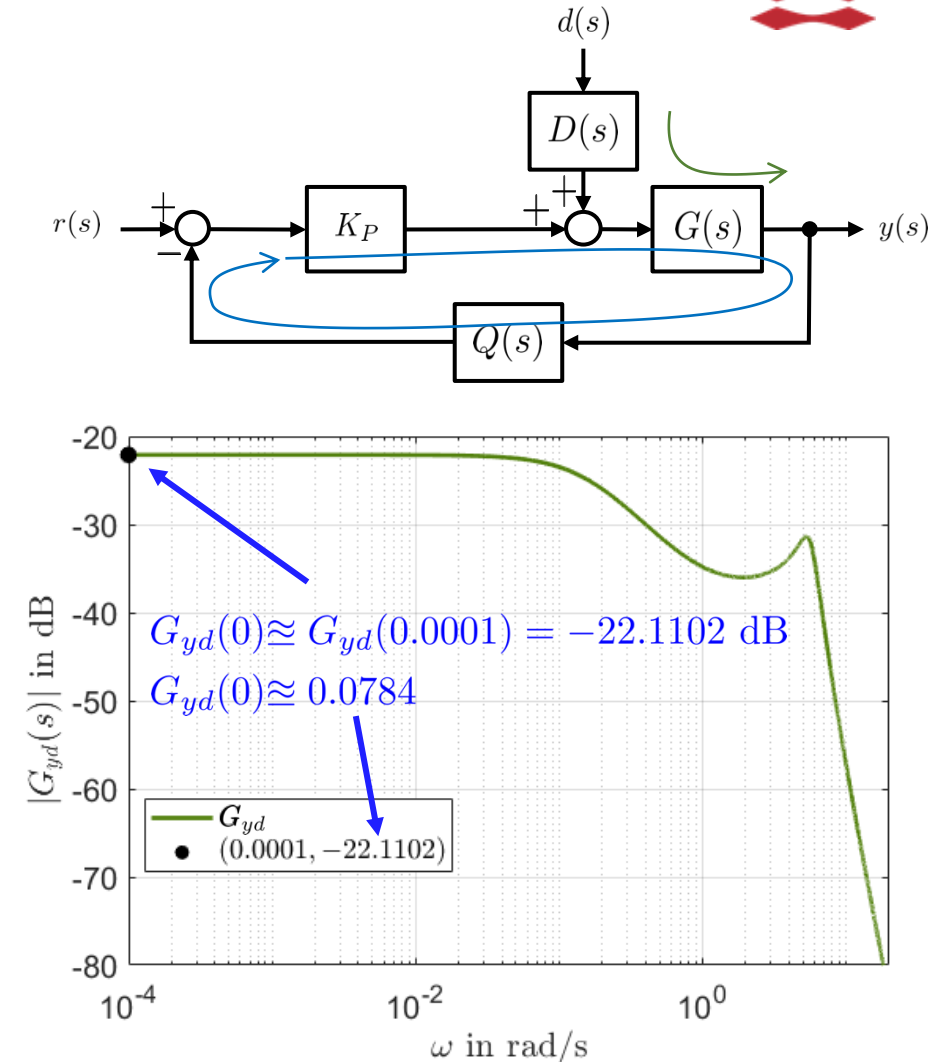
Example (Exam 2023)

Consider the following block diagram where $G(0) = 10$, $Q(0) = 1$ and $D(0) = 0.4$. The magnitude Bode plot of the sensitivity transfer function $G_{yd}(s)$ of the output to the disturbance $d(s)$ is shown in the figure below. What is (approximately) the value of the proportional gain K_P ?

Hint: 0.0001 rad/s can be considered as "steady-state".

- a) $K_P = -0.1181$
- b) $K_P = 5$
- c) $K_P = -5.2$
- d) $K_P = 2$
- e) $K_P = 0.32$

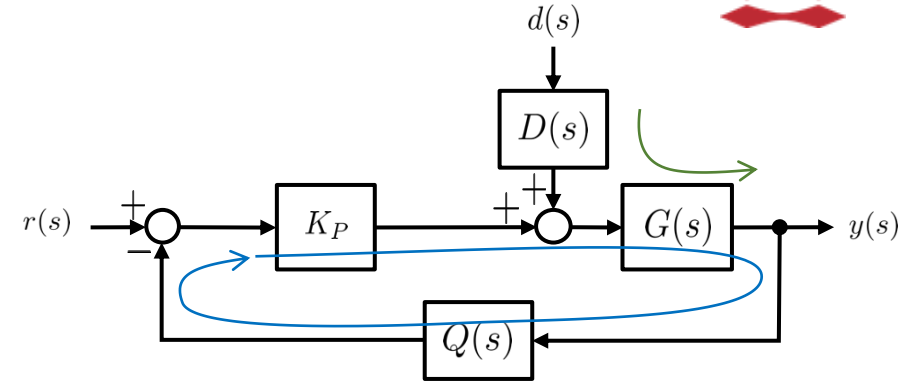
$$G_{yd}(s) = \frac{D(s)G(s)}{1 + K_P G(s)Q(s)}$$



Sensitivity

Example (Exam 2023)

Consider the following block diagram where $G(0) = 10$, $Q(0) = 1$ and $D(0) = 0.4$. The magnitude Bode plot of the sensitivity transfer function $G_{yd}(s)$ of the output to the disturbance $d(s)$ is shown in the figure below. What is (approximately) the value of the proportional gain K_P ?



Hint: 0.0001 rad/s can be considered as "steady-state".

- a) $K_P = -0.1181$ Used $G_{yd}(0)$ in dB
- b) $K_P = 5$ Correct
- c) $K_P = -5.2$ Wrong calculation of $G_{yd}(s)$
- d) $K_P = 2$ Wrong calculation of $G_{yd}(s)$
- e) $K_P = 0.32$ Wrong calculation of $G_{yd}(s)$

$$G_{yd}(s) = \frac{D(s)G(s)}{1 + K_P G(s)Q(s)} \xrightarrow{G_{yd}(0) \approx 0.0784}$$

$$0.0784 \approx \frac{D(0)G(0)}{1 + K_P G(0)Q(0)} = \frac{4}{1 + 10K_P} \Rightarrow$$

$$K_P \approx \frac{1}{10} \left(\frac{4}{0.0784} - 1 \right) = 5$$

Pre-filters

Study case: $G_{cl}(s) = \frac{13.1s^3 + 209.6s^2 + 851.5s + 655}{s^4 + 10s^3 + 156s^2 + 1036s + 655}$

- Resonance Peak → oscillations.** A Lead part could help but it may amplify the noise of the measurements.

- A pre-filter could achieve dampening of the oscillations.

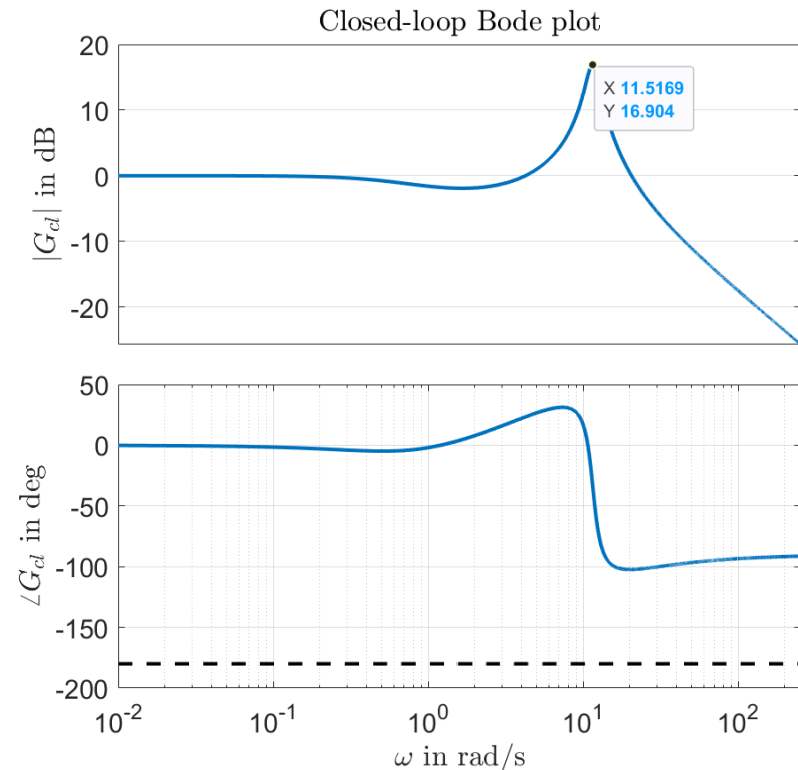
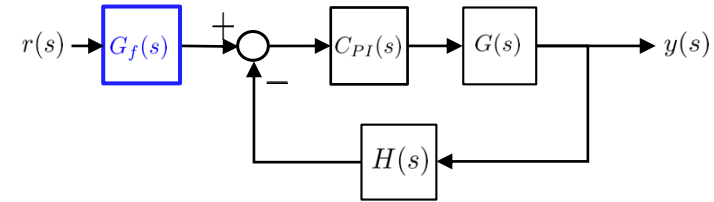
$$G_f(s) = \frac{1}{\tau_f s + 1}$$

- At peak frequency $\omega_p = 11.52$ rad/sec the magnitude of G_{cl} is 16.9 dB or $|G_{cl}(j\omega_p)| = 7$.

- We require that the peak in the filtered closed-loop is reduced to 0 dB.

$$\left| \frac{1}{\tau_f \omega_p j + 1} G_{cl}(j\omega_p) \right| = 1 \Leftrightarrow |\tau_f 11.52j + 1| = 7$$

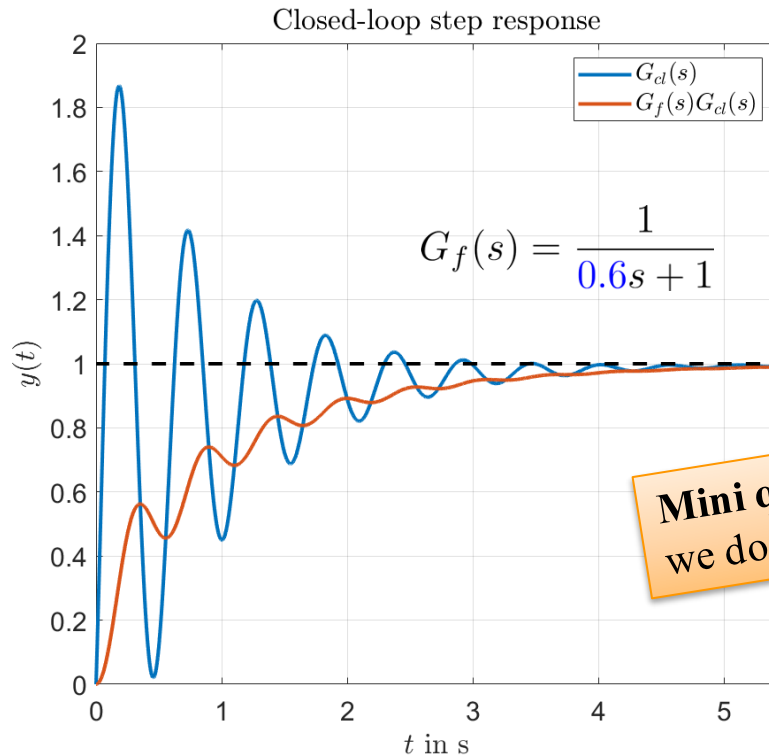
$$\Leftrightarrow (\tau_f 11.52)^2 + 1 = 7^2 \Rightarrow \tau_f \approx 0.6$$



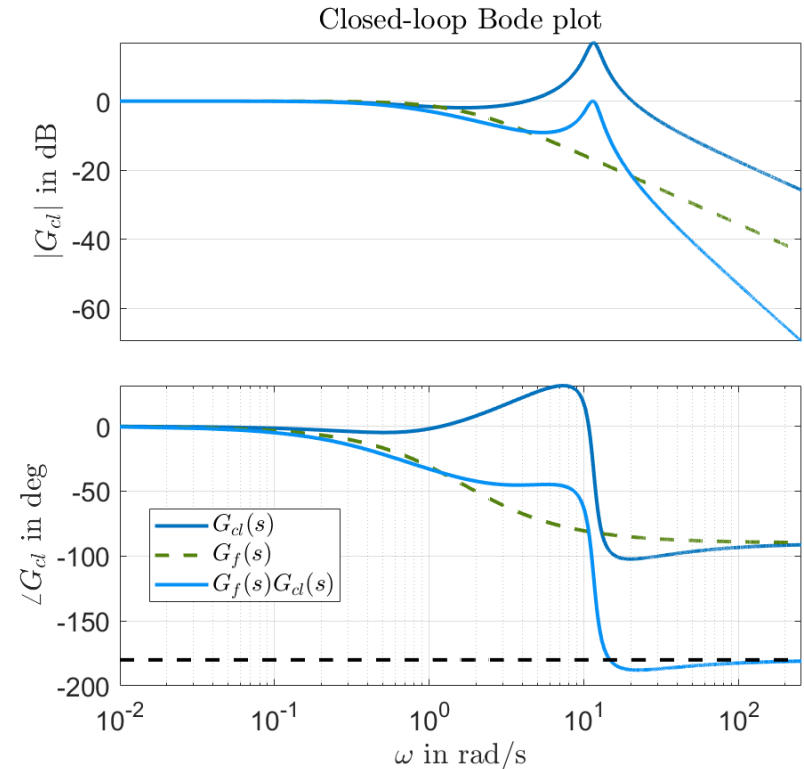
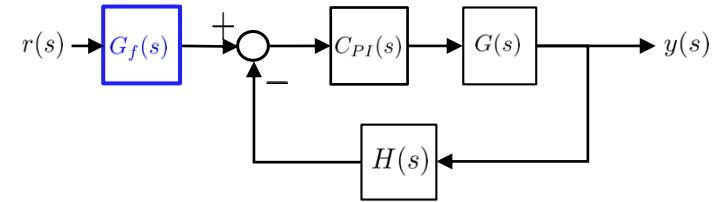
Pre-filters

Study case: $G_{cl}(s) = \frac{13.1s^3 + 209.6s^2 + 851.5s + 655}{s^4 + 10s^3 + 156s^2 + 1036s + 655}$

- A pre-filter could achieve dampening of the oscillations.



Mini quiz: Can we do better?



Pre-filters

Study case: $G_{cl}(s) = \frac{13.1s^3 + 209.6s^2 + 851.5s + 655}{s^4 + 10s^3 + 156s^2 + 1036s + 655}$

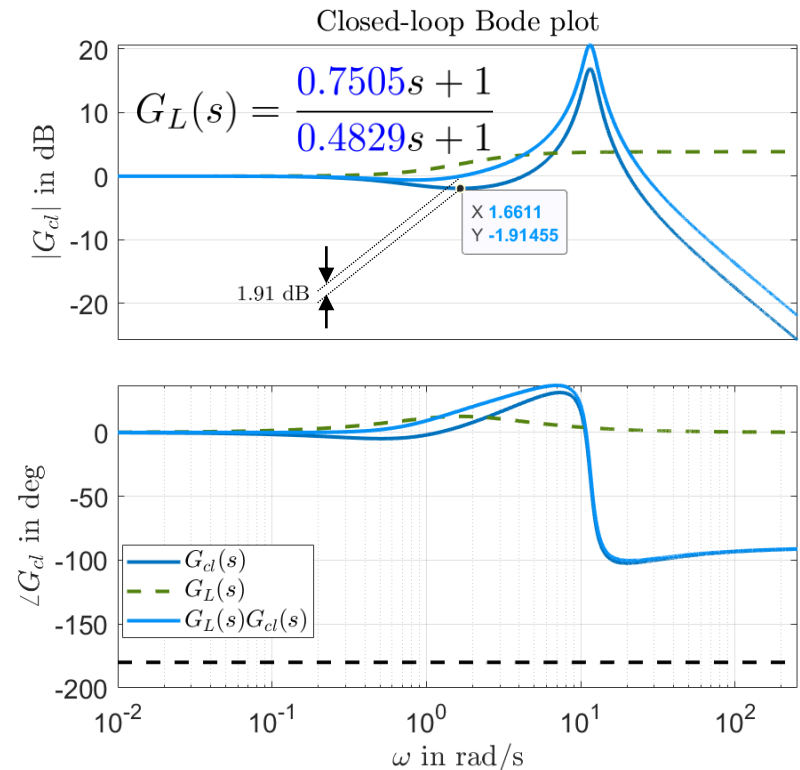
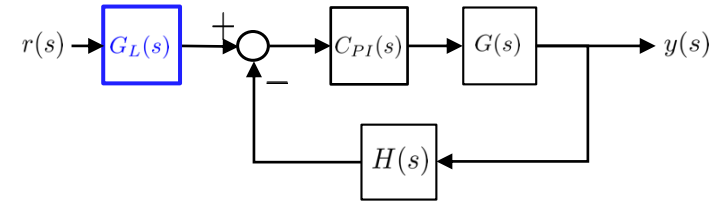
- The “valley” before the peak can be smoothened out with a Lead part (as pre-filter now).

$$G_L(s) = \frac{\tau_d s + 1}{\alpha \tau_d s + 1}$$

- The deepest point of the “valley” is at $\omega_v = 1.66$ rad/sec, where $|G_{cl}(j\omega_v)| = -1.91$ dB = 0.8 .
- We select α such that the Lead part give its known magnitude contribution there and we require that:

$$|G_L(j\omega_v)G_{cl}(j\omega_v)| = 1 \Leftrightarrow \frac{1}{\sqrt{\alpha}} = \frac{1}{0.8} \Rightarrow \alpha \approx 0.6435$$

$$\tau_d = \frac{1}{\omega_v \sqrt{\alpha}} \Rightarrow \tau_d \approx 0.7505$$



Pre-filters

Study case: $G_{cl}(s) = \frac{13.1s^3 + 209.6s^2 + 851.5s + 655}{s^4 + 10s^3 + 156s^2 + 1036s + 655}$

- We still need to deal with the peak.
 - Caused by a pair of complex conjugate poles
 - IDEA: introduce a pair of complex conjugate zeros to counteract the poles.

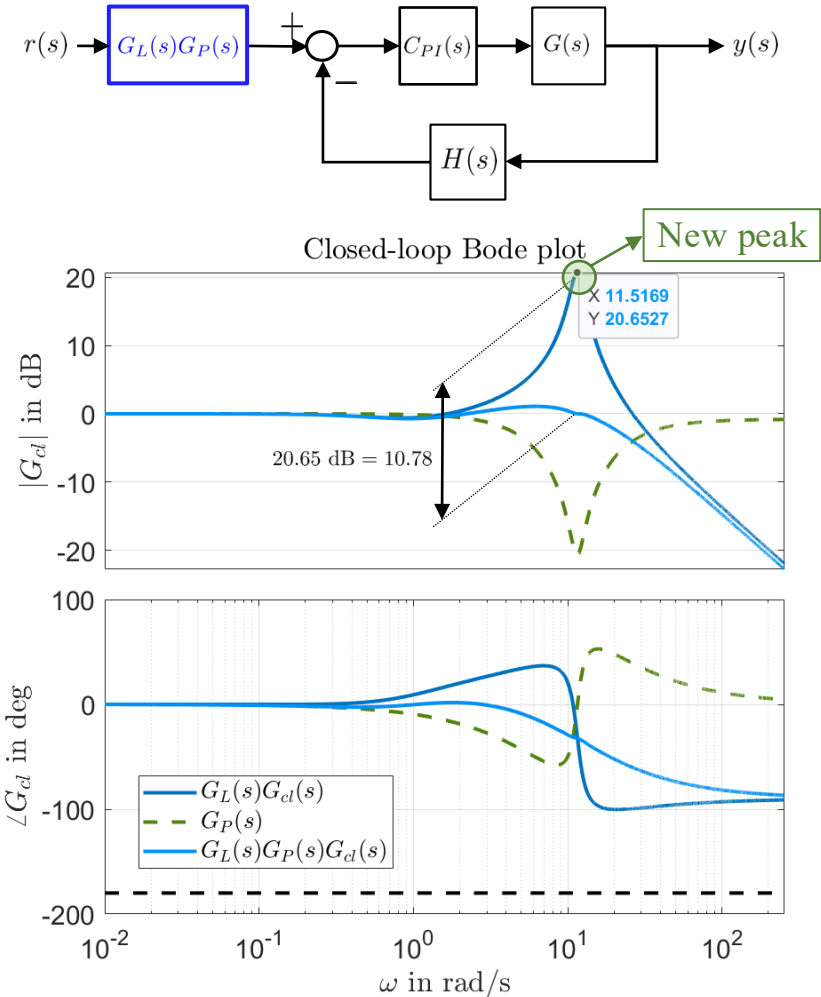
$$G_P(s) = \frac{s^2 + 2\zeta\omega_p s + \omega_p^2}{(s + p)^2} \cdot \frac{p^2}{\omega_p^2}$$

- The double real pole p is chosen a bit smaller (faster) than the fastest zero of $G_{cl}(s)$, here $p = 11$.

- To find ζ , we require that

$$|G_L(\omega_p j)G_P(\omega_p j)G_{cl}(j\omega_p)| = 1 \Leftrightarrow$$

$$\frac{2\zeta p^2}{\sqrt{(p^2 - \omega_p^2)^2 + 4\omega_p^2 p^2}} = \frac{1}{10.78} \Rightarrow \zeta \approx 0.0972$$

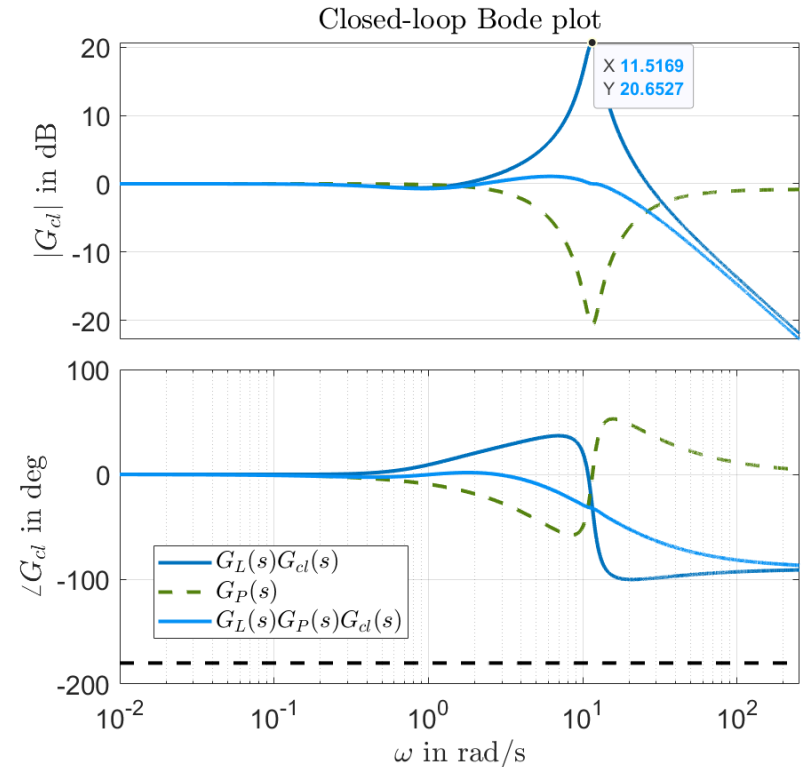
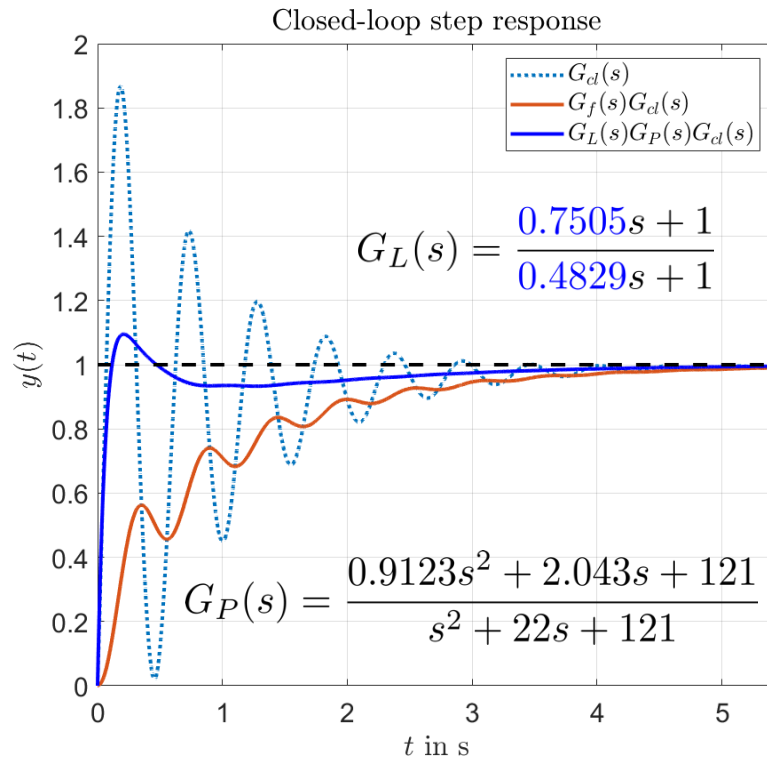
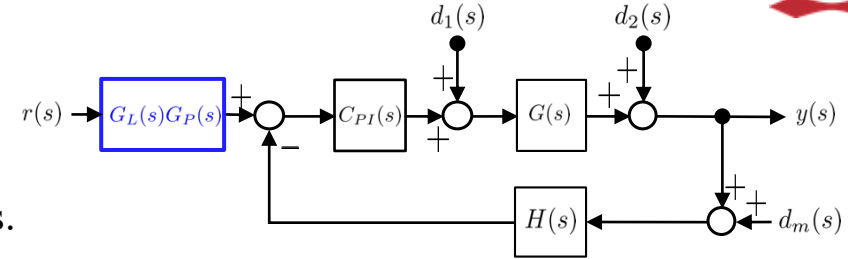


Pre-filters

prefilters.m

Study case: $G_{cl}(s) = \frac{13.1s^3 + 209.6s^2 + 851.5s + 655}{s^4 + 10s^3 + 156s^2 + 1036s + 655}$

- A pre-filter can achieve dampening of the oscillations.



Today you learned

- Disturbances and Sensitivity
 1. How to find the transfer function from a disturbance to the output/error.
 2. How to calculate the steady-state error due to a disturbance.
 3. What the sensitivity of system output and control input to disturbances is.
 4. How to evaluate the effect of disturbances on the output and input of a system from the Bode plot.
- Pre-filters
 1. How to use pre-filters in the reference.
 2. How to design first and second-order pre-filters.

Next lecture 13

Feed-forward control

- Design feed-forward controllers for improving tracking performance.
- Design feed-forward controllers for disturbance rejection.

Today's Exercises

Group work: Continue on REGBOT balance (mandatory assignment)

Home alone: Continue on the theoretical exercises LCD1: 1, 2, 3, 4, 5, 6, 7, 8, 9